

The main question I am addressing through this experiment is to explore the effect of interactions between thermal and saline stress to determine the capacity of starlet sea anemones for plasticity. These animals are in estuarine habitats that are typically very dynamic, so we expect the anemones to be able to acclimate to various stressors relatively easily. In the context of my thesis, I want to first outline specific acclimation responses to thermal and saline stress before characterising them further.

As a physiologist, I am interested in metabolic responses, so I have measured the routine metabolic rate (RMR). I have also measured/plan to measure some enzyme activities – citrate synthase (CS) (mitochondrial marker), ATPases (osmoregulatory enzymes), and some enzymes involved in anaerobic respiration. I am also measuring some markers of oxidative stress on lipids and proteins (TBARS, PCC). These should give me an idea of where compensation broadly occurs, if any. The differences in aerobic and anaerobic respiration helps with understanding changes in energy budgeting. One of the osmoregulatory enzymes I am measuring is the $\text{Na}^+\text{-K}^+$ -ATPase (NKA), which is energetically expensive and can thus explain osmoregulatory costs (if any).

The acclimation treatments (high or low temperature and high or low salinity) are both nominal variables. They could also be an interval, 0°C is a real temperature and 0ppt is a real salinity. I think I would treat them as nominal because I am characterising these values are ‘high’ or ‘low’, I am not studying them at other temperature or salinity ranges to provide that context.

I am using repeated measures from eight different genotypes. This variable is a nominal type; the order should not matter - although I have never randomly ordered the genotypes during experiments and I should have. I have followed the exact same order for all experiments when I load a plate for example, so this could be considered ordinal as well.

I have measured the masses of all anemones before running any of the experiments, the masses are continuous and have a meaningful zero, so they are on the ratio scale.

I plan to analyse all the above variables (RMR, CS, PCC, TBARS, NKA, ATPases, anaerobic enzymes) using the log scale. These are all continuous variables and have a meaningful zero and thus are also of the ratio scale type. I prefer the log scale because I don’t have information on the allometric relationship between these variables with mass, and therefore, I plan to just assume a linear relationship on the log scale ($\log \text{RMR} \sim \log \text{mass}$, for example). Further, these measures are small and can have more resolution on the log scale.

I have looked at other studies that have measured these criteria in the anemones or in other cnidarians (Sachkova *et al.*, 2020; Rivera *et al.*, 2021; Angonese *et al.*, 2022; Coll *et al.*, 2025; Glass & Barott, 2025). I have checked that my values are within range. I have no way of knowing what a meaningful effect size is, but I can compare effects of acclimation to these stressors to

other benthic animals (my lab has some data on a similar experimental design but using killifish) just to provide context. I also plan to model the data and calculate estimated marginal means with confidence intervals to realistically analyse/present it.

A major motivation for this experiment was to look for quantifiable responses to these stressors. For my later studies in my PhD, I am interested in determining mechanisms/processes underlying inter- and intra-individual variation in the capacity for plasticity by leveraging the clonal properties of *Nematostella vectensis* as a model organism. However, very little is known about plastic responses of *Nematostella* to various stressors. My plan thus far has therefore been to first identify and outline the acclimation responses in starlet sea anemones to ecologically relevant stressors.

References

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